

Gesture Recognition System

Overview

This project focuses on building a **Gesture Recognition System** using Machine Learning / Deep Learning techniques to identify and classify human gestures from images or video sequences.

Gesture recognition plays a crucial role in **human-computer interaction (HCI)**, enabling touchless control systems used in applications like gaming, virtual reality, sign language recognition, and smart interfaces.

Problem Statement

The goal of this project is to:

- Detect and classify hand/body gestures accurately
 - Enable real-time interaction between humans and machines
 - Improve accessibility through gesture-based controls
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Dataset Description

The dataset consists of gesture images or video frames categorized into different classes such as:

- Hand movements (left, right, up, down)
 - Static gestures (stop, thumbs up, etc.)
 - Dynamic gestures (wave, swipe)
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Data Preprocessing

- Image resizing and normalization
 - Noise reduction
 - Frame extraction (for video-based input)
 - Data augmentation (rotation, flipping, scaling)
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Feature Engineering

- Keypoint detection (optional using OpenCV / MediaPipe)
- Image feature extraction
- Sequence modeling for temporal gestures



Models Implemented

Machine Learning (if applicable)

- Support Vector Machine (SVM)
- Random Forest

Deep Learning

- Convolutional Neural Networks (CNN) for image-based gestures
- LSTM / GRU for sequence-based gesture recognition
- CNN + LSTM hybrid model for video gesture classification



Evaluation Metrics

- Accuracy
- Precision, Recall, F1-score
- Confusion Matrix



Key Insights

- CNN models perform well for static gesture recognition
- LSTM models improve accuracy for dynamic gestures
- Data augmentation significantly improves generalization
- Real-time prediction depends heavily on preprocessing speed



Tech Stack

- Python
- OpenCV
- TensorFlow / Keras / PyTorch
- NumPy, Pandas
- MediaPipe (for hand tracking)



Applications

- Virtual mouse / keyboard
- Sign language recognition
- Smart home control
- Gaming interfaces

- AR/VR interaction systems
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Future Work

- Improve real-time detection performance
- Deploy model using Flask / FastAPI
- Integrate with mobile or edge devices
- Expand dataset for better accuracy